Nicholas J. Seewald, Ph.D.
Curriculum Vitae

seewaldn@pennmedicine.upenn.edu nickseewald.com nickseewald

Professional Data

Department of Biostatistics, Epidemiology, and Informatics
University of Pennsylvania Perelman School of Medicine
3600 Civic Center Drive
Room 3W-103
Philadelphia, PA 19104

Education

University of Michigan, Ann Arbor, MI.

Ph.D. Statistics 2021

Design and analytic considerations for sequential, multiple-assignment randomized trials with continuous longitudinal outcomes. Advised by Daniel Almirall, Ph.D. doi.org/n4r2

M.A. Statistics 2018

M.S. Biostatistics 2015

University of Notre Dame, Notre Dame, IN.

B.S. Mathematics (with Life Science), *cum laude* 2013

Faculty & Research Positions

University of Pennsylvania, Philadelphia, PA.

Senior Scholar, Center for Clinical Epidemiology and Biostatistics 2025 - Present

Assistant Professor of Biostatistics 2023 - Present

Johns Hopkins Bloomberg School of Public Health, Baltimore, MD.

Postdoctoral Fellow, Department of Health Policy and Management 2021-2023
Advised by Elizabeth A. Stuart, Ph.D. and Emma E. (Beth) McGinty, Ph.D.

Awards

8. Chair's Junior Faculty Spotlight Award. Department of Biostatistics, Epidemiology, and Informatics, University of Pennsylvania Perelman School of Medicine, 2026.

7. Excellence in Teaching. Johns Hopkins Bloomberg School of Public Health, AY 2022-23, First Term.

6. Junior Researcher Travel Grant, American Causal Inference Conference, 2022. Supported by the National Science Foundation, awarded to Duke University.

5. Brenda Gunderson Graduate Student Instructor Award. Department of Statistics, University of Michigan, 2021. (Inaugural recipient)

4. Outstanding Graduate Student Instructor. Rackham Graduate School, University of Michigan, 2021. (University-wide)

3. Outstanding Graduate Student Instructor. Department of Statistics, University of Michigan, 2020.

2. Best Departmental Poster, Statistics. Michigan Student Symposium for Interdisciplinary Statistical Sciences, 2017.

1. Best Departmental Poster, Biostatistics. Michigan Student Symposium for Interdisciplinary Statistical Sciences, 2015.

Publications

* denotes trainee with significant mentoring.

Peer-Reviewed Articles

29. Mayer M, **Seewald NJ**, Perez EU, Moreno BH, Ramamurthy C, Babcock A, Li H, and Mamtani R, "Real-world treatment duration with first-line enfortumab vedotin plus pembrolizumab for advanced urothelial cancer: A national electronic health record study," *Clinical Genitourinary Cancer*, p. 102 652, 2026. DOI: 10.1016/j.clgc.2026.102652

28. *Gerety MK, Charu V, Simmons M, **Seewald NJ**, Glenn DA, Schaubel DE, Smith AR, Chen D, Laurin LP, Denburg M, Holzman LB, Beck LH, Mariani LH, and Zee J, "Real-world long-term comparisons of rituximab versus calcineurin inhibitors for membranous nephropathy in the cure glomerulonephropathy study," *Kidney360*, 2026. DOI: 10.34067/KID.0000001265

27. Samuel LJ, Wright RS, Charankevich H, **Seewald NJ**, Tian J, Dwivedi P, and Szanton SL, "Health effects of an increased federal minimum wage among US low-wage-earning adults: A difference-in-differences analysis in a population-based cohort," *BMJ Public Health*, vol. 4, no. 3, 2026. DOI: 10.1136/bmjph-2025-003487 PMID: 10.1136/bmjph-2025-003487.

26. Eisenberg MD, Tormohlen KN, McCourt AD, Yu J, Neupane R, Ge Y, Stuart EA, Fingerhood M, **Seewald NJ**, and McGinty EE, "Impact of state telemedicine policies on substance use disorder treatment during the COVID-19 pandemic," *Journal of General Internal Medicine*, 2026. DOI: 10.1007/s11606-026-10394-7

25. O'Leary C, Tischfield DJ, Avritscher R, El-Haddad G, Garcia-Monaco R, Fidelman N, Vijayvergia N, Kunz PL, **Seewald NJ**, and Soulen MC, "Associations between patterns of neuroendocrine liver metastatic burden and outcomes after liver-directed therapy, systemic chemotherapy, and peptide receptor radionuclide therapy," *Neuroendocrinology*, 2026. DOI: 10.1159/000551977

24. *Moin EE, Bennett RM, Moffett AT, Schmid BE, Long K, **Seewald NJ**, Reilly JP, and Weissman GE, "Waveform data capture substantial variation in Tidal volume and other respiratory parameters," *Annals of the American Thoracic Society*, 2026. DOI: 10.1093/annalsats/aaoag024

23. Till JE, Gal-Rosenberg O, Giliberto SG, **Seewald NJ**, Ballinger DG, Samberg HE, Yin MR, Wang QL, Cannas S, Kim KN, Tien K, Sawi M, Madineedi V, Furniss CS, Gocheva V, Nowak J, Brais LK, Yuan C, Rosenthal MH, Roses R, DeMatteo R, Lee MK, Vollmer C, Sagreiya H, O'Hara MH, Shemer R, Wolpin B, Dor Y, and Carpenter EL, "Plasma cell-free DNA markers predict occult metastases in patients with resectable pancreatic ductal adenocarcinoma," *Clinical and Translational Medicine*, vol. 16, no. 1, e70573, 2026. DOI: 10.1002/ctm2.70573

22. Stone EM, **Seewald NJ**, and Rosen JG, "Impact of medicaid expansion on HIV pre-exposure prophylaxis coverage, 2012–23," *Health Affairs*, vol. 44, no. 10, pp. 1266–1272, 2025. DOI: 10.1377/hlthaff.2025.00211

21. McGinty EE, Wagle P, Luo CL, **Seewald NJ**, Stuart EA, and Tormohlen KN, "The impact of medical cannabis laws on cannabis and opioid use disorder treatment and overdose-related health care utilization among adults with chronic noncancer pain," *Milbank Quarterly*, vol. 103, no. S1, pp. 411–434, 2025. DOI: 10.1111/1468-0009.70052

20. Anbil S, **Seewald NJ**, Chiorean EG, Hussein M, Kasi PM, Laux DE, Schwartz GK, Shapiro GI, Lin KK, Craib M, Maloney L, McLachlan K, Tukachinsky H, Schrock AB, Wang S, Sokol ES, Decker B, Nathanson KL, Domchek SM, and Reiss KA, "LODESTAR: A single-arm phase II study of rucaparib in solid tumors with pathogenic germline or somatic variants in homologous recombination repair genes," *JCO Precision Oncology*, no. 9, e2500090, 2025. DOI: 10.1200/PO-25-00090

19. Hamm RF, Mumford SL, Forkpa M, **Seewald NJ**, Beidas RS, Srinivas SK, Parry S, and Levine LD, "The impact of implementing a standardized protocol for labor induction on obstetric disparities: Secondary analysis of a type I hybrid effectiveness-implementation trial," *Pregnancy*, vol. 1, no. 4, e70077, 2025. DOI: 10.1002/pmf2.70077

18. *Moin EE, **Seewald NJ**, and Halpern SD, "Use of life support and outcomes among patients admitted to intensive care units," *JAMA*, vol. 333, no. 20, pp. 1793–1803, 2025. DOI: 10.1001/jama.2025.2163

17. Till JE, **Seewald NJ**, Yazdani Z, Wang Z, Ballinger D, Samberg H, Dandu S, Macia C, Yin M, Abdalla A, Prior T, Shah SS, Patel T, McCoy E, Mansour M, Wills CA, Bochenek V, Serrano J, Snuderl M, Phillips RE, O'Rourke DM, Amankulor NM, Nabavizadeh A, Desai AS, Gollomp K, Binder ZA, Zhou W, Bagley SJ, and Carpenter EL, "Corticosteroid-dependent association between prognostic peripheral blood cell-free DNA levels and neutrophil-mediated NETosis in patients with glioblastoma," *Clinical Cancer Research*, vol. 31, no. 7, pp. 1292–1304, 2025. DOI: 10.1158/1078-0432.CCR-24-3169

16. Stone EM, Jopson AD, **Seewald NJ**, Stuart EA, Wise E, McCourt AD, German D, and McGinty EE, "Effects of texas state agency integration on mental health service use among individuals with co-occurring cognitive disabilities and mental health conditions," *Community Mental Health Journal*, vol. 61, pp. 111–121, 2025. DOI: 10.1007/s10597-024-01332-0

15. **Seewald NJ**, McGinty EE, and Stuart EA, "Target trial emulation for evaluating health policy," *Annals of Internal Medicine*, 2024. DOI: 10.7326/M23-2440

14. McGinty EE, Tormohlen KN, **Seewald NJ**, Bicket MC, McCourt AD, Rutkow L, White SA, and Stuart EA, "Effects of U.S. state medical cannabis laws on treatment of chronic noncancer pain," *Annals of Internal Medicine*, vol. 176, no. 7, pp. 904–912, 2023. DOI: 10.7326/M23-0053

13. McGinty EE, **Seewald NJ**, Bandara S, Cerdá M, Daumit GL, Eisenberg MD, Griffin BA, Igusa T, Jackson JW, Kennedy-Hendricks A, Marsteller J, Miech EJ, Purtle J, Schmid I, Schuler MS, Yuan CT, and Stuart EA, "Scaling interventions to manage chronic disease: Innovative methods at the intersection of health policy research and implementation science," *Prevention Science*, 2022. DOI: 10.1007/s11121-022-01427-8

12. McGinty EE, Bicket MC, **Seewald NJ**, Stuart EA, Alexander GC, Barry CL, McCourt AD, and Rutkow L, "Effects of state opioid prescribing laws on use of opioid and other pain treatments among commercially insured U.S. adults," *Annals of Internal Medicine*, vol. 175, no. 5, pp. 617–627, 2022. DOI: 10.7326/M21-4363

11. **Seewald NJ**, Kidwell KM, Nahum-Shani I, Wu T, McKay JR, and Almirall D, "Sample size considerations for comparing dynamic treatment regimens in a sequential multiple-assignment randomized trial with a continuous longitudinal outcome," *Statistical Methods in Medical Research*, vol. 29, no. 7, pp. 1891–1912, 2020. DOI: 10/gf85ss
10. **Seewald NJ**, Smith SN, Lee AJ, Klasnja P, and Murphy SA, "Practical considerations for data collection and management in mobile health micro-randomized trials," *Statistics in Biosciences*, vol. 11, pp. 355–370, 2019. DOI: 10/gfsvx7
9. Kidwell KM, **Seewald NJ**, Tran Q, Kasari C, and Almirall D, "Design and analysis considerations for comparing dynamic treatment regimens with binary outcomes from sequential multiple assignment randomized trials," *Journal of Applied Statistics*, vol. 45, no. 9, pp. 1628–1651, 2018. DOI: 10/hh5r
8. Klasnja P, Smith S, **Seewald NJ**, Lee A, Hall K, Luers B, Hekler EB, and Murphy SA, "Efficacy of contextually tailored suggestions for physical activity: A micro-randomized optimization trial of HeartSteps," *Annals of Behavioral Medicine*, p. 10, 2018. DOI: 10.1093/abm/kay067
7. Kadakia KC, Kidwell KM, **Seewald NJ**, Snyder CF, Storniolo AM, Otte JL, Flockhart DA, Hayes DF, Stearns V, and Henry NL, "Prospective assessment of patient-reported outcomes and estradiol and drug concentrations in patients experiencing toxicity from adjuvant aromatase inhibitors," *Breast Cancer Research and Treatment*, vol. 164, no. 2, pp. 411–419, 2017. DOI: 10.1007/S10549-017-4260-2
6. Meurer WJ, **Seewald NJ**, and Kidwell K, "Sequential multiple assignment randomized trials: An opportunity for improved design of stroke reperfusion trials," *Journal of Stroke and Cerebrovascular Diseases*, vol. 26, no. 4, pp. 717–724, 2017. DOI: 10.1016/j.jstrokecerebrovasdis.2016.09.010
5. Kadakia KC, Snyder CF, Kidwell KM, **Seewald NJ**, Flockhart DA, Skaar TC, Desta Z, Rae JM, Otte JL, Carpenter JS, Storniolo AM, Hayes DF, Stearns V, and Henry NL, "Patient-reported outcomes and early discontinuation in aromatase inhibitor-treated postmenopausal women with early stage breast cancer," *The Oncologist*, vol. 21, no. 5, pp. 539–546, 2016. DOI: 10.1634/theoncologist.2015-0349 PMID: 27009936.
4. Hertz DL, Caram MV, Kidwell KM, Thibert JN, Gersch C, **Seewald NJ**, Smerage J, Rubenfire M, Henry NL, Cooney KA, Leja M, Griggs JJ, and Rae JM, "Evidence for association of SNPs in ABCB1 and CBR3, but not RAC2, NCF4, SLC28A3 or TOP2B, with chronic cardiotoxicity in a cohort of breast cancer patients treated with anthracyclines.," *Pharmacogenomics*, vol. 17, no. 3, pp. 231–240, 2016. DOI: 10.2217/pgs.15.162 PMID: 26799497.
3. Hertz DL, Kidwell KM, **Seewald NJ**, Gersch CL, Desta Z, Flockhart DA, Storniolo AM, Stearns V, Skaar TC, Hayes DF, Henry NL, and Rae JM, "Polymorphisms in drug-metabolizing enzymes and steady-state exemestane concentration in postmenopausal patients with breast cancer," *The Pharmacogenomics Journal*, pp. 1–7, November 2015 2016. DOI: 10.1038/tpj.2016.60
2. Randolph AH, **Seewald NJ**, Rickert K, and Brown SN, "Tris (3,5-di-tert-butylcatecholato) molybdenum(VI): Lewis acidity and nonclassical oxygen atom transfer reactions.," *Inorganic chemistry*, vol. 52, no. 21, pp. 12 587–98, 2013. DOI: 10.1021/ic401736f PMID: 24147870.
1. Marshall-Roth T, Liebscher SC, Rickert K, **Seewald NJ**, Oliver AG, and Brown SN, "Nonclassical oxygen atom transfer reactions of oxomolybdenum(VI) bis(catecholate)," *Chemical Communications*, vol. 48, no. 63, pp. 7826–7828, 2012. DOI: 10.1039/c2cc33523a PMID: 22785616.

Peer-Reviewed Book Chapters

3. Smith SN, **Seewald NJ**, and Klasnja P, "Design considerations for preparation, optimization, and evaluation of digital therapeutics," *Digital Therapeutics for Mental Health and Addiction*, Jacobson N, Marsch L, and Kowatsch T, Eds., Academic Press, 2023, pp. 135–150. DOI: 10.1016/B978-0-323-90045-4.00015-0
2. **Seewald NJ**, Hackworth O, and Almirall D, "Sequential, multiple assignment, randomized trials (SMART)," *Principles and Practice of Clinical Trials*, Piantadosi S and Meinert CL, Eds., Cham: Springer International Publishing, 2021, pp. 1–19. DOI: 10.1007/978-3-319-52677-5_280-1
1. Smith SN, Lee AJ, Hall K, **Seewald NJ**, Boruvka A, Murphy SA, and Klasnja P, "Design lessons from a micro-randomized pilot study in mobile health," *Mobile Health*, Rehg JM, Murphy SA, and Kumar S, Eds., Cham: Springer International Publishing, 2017, pp. 59–82. DOI: 10.1007/978-3-319-51394-2_4

Abstracts

15. *Mayer M, **Seewald N**, Babcock A, Homet Moreno B, Ramamurthy C, Li H, and Mamtani R, "Untreated advanced urothelial carcinoma (aUC) in the modern treatment era: Patterns of care from a decade-long national cohort.," *Journal of Clinical Oncology*, vol. 44, 2026, pp. 4570–4570. DOI: 10.1200/JCO.2026.44.16_suppl.4570
14. Montgomery AM, Vachani A, Saia C, Kalman J, Singletary P, Vecchio ND, Peifer M, Wender R, Gabriel P, **Seewald N**, Wainwright J, Toneff H, Steltz J, Zeb S, and Rendle KA, "Contextual Factors Influencing the Effectiveness of Multilevel Strategies on Annual Lung Cancer Screening and Diagnostic Follow-up," *American Journal of Respiratory and Critical Care Medicine*, (Orlando, FL), vol. 212, 2026, A100–06. DOI: 10.1093/ajrccm/aamag162.3773
13. Zeb S, Steltz J, Saia C, Peifer M, Wender R, Gabriel P, **Seewald N**, Vachani A, and Rendle K, "Increasing Equitable Adherence to Annual Lung Cancer Screenings and Diagnostic Surveillance: A Pragmatic Factorial Trial," *Journal of the National Comprehensive Cancer Network*, (Orlando, FL, USA), vol. 24, National Comprehensive Cancer Network, 2026, QIM26–279. DOI: 10.6004/jnccn.2025.7182
12. *Taranto E, **Seewald NJ**, Bayne LJ, Walinsky E, Deluca S, Shih NN, Sanchez P, Nivar I, Ambasager B, Murray C, Chevalier A, Smith CG, Makhlin I, Rohn K, Goodspeed BL, Savage J, Wileyto P, Wang J, Belka G, Chislock E, Berry LR, Berry D, Nayak A, Feldman M, Clark AS, Chodosh LA, and DeMichele A, "Circulating tumor DNA (ctDNA), dormant disseminated tumor cells (DTCs) and recurrence outcomes in breast cancer survivors on the SURMOUNT Study," *Clinical Cancer Research*, vol. 31, San Antonio, TX: AACR, 2025, PS9–03. DOI: 10.1158/1557-3265.SABCS24-PS9-03
11. *Moin EE, **Seewald NJ**, and Halpern SD, "Characteristics and prognosis of patients experiencing recurrent ICU admissions," *American Journal of Respiratory and Critical Care Medicine*, American Thoracic Society, 2025. DOI: 10.1164/ajrccm.2025.211.Abstracts.A7371
10. *Moin EE, **Seewald NJ**, Halpern SD, and Auriemma CL, "Measurement error in hospital-free days after prolonged acute mechanical ventilation," *American Journal of Respiratory and Critical Care Medicine*, American Thoracic Society, 2025. DOI: 10.1164/ajrccm.2025.211.Abstracts.A7792

9. O'Leary C, Tischfield D, Avritscher R, El-Haddad G, Fidelman N, Garcia-Monaco R, Vijayvergia N, Kunz P, **Seewald N**, and Soulen MC, "Does liver tumor morphology predict outcomes after LDT, PRRT or captem?" *Endocrine Abstracts*, vol. 108, Chicago, IL: Bioscientifica, 2025. DOI: 10.1530/endoabs.108.C25
8. *Mayer M, **Seewald N**, Perez EU, Homet Moreno B, Ramamurthy C, Babcock A, Li H, and Mamtani R, "Real-world time on treatment (rwTOT) with first-line (1L) enfortumab vedotin and pembrolizumab (EV+P) after U.S. Food and Drug Administration approval for advanced urothelial cancer (aUC).," *Journal of Clinical Oncology*, vol. 43, Wolters Kluwer, 2025, pp. 731–731. DOI: 10.1200/JCO.2025.43.5_suppl.731
7. Anbil S, **Seewald NJ**, Lin K, Craib M, Connor C, Giordano H, McLachlan K, Kwan T, Maloney L, Tukachinsky H, Schrock A, Wang S, Sokol E, Decker B, Nathanson K, Domchek S, and Reiss K, "LODESTAR: A single arm phase II study of rucaparib in solid tumors with pathogenic germline or somatic variants in homologous recombination repair (HRR) genes," *Cancer Research*, vol. 84, Boston, MA: AACR, 2024, p. C006. DOI: 10.1158/1538-7445.PANCREATIC24-C006
6. Hood R, Rathore S, **Seewald NJ**, and Reiss KA, "Association of BRCA1/2 pathogenic variants with primary tumor location and metastatic organotropism in pancreatic adenocarcinoma.," *Journal of Clinical Oncology*, vol. 42, Chicago, IL: Wolters Kluwer, 2024, pp. 4151–4151. DOI: 10.1200/JCO.2024.42.16_suppl.4151
5. Hood R, **Seewald NJ**, Rathore S, and Reiss KA, "Interaction of BRCA1 vs BRCA2 pathogenic variants with primary tumor location in pancreatic cancer.," *Journal of Clinical Oncology*, vol. 42, Chicago, IL: Wolters Kluwer, 2024, e16326–e16326. DOI: 10.1200/JCO.2024.42.16_suppl.e16326
4. Hertz DL, Kidwell KM, **Seewald NJ**, Gersch CL, Desta Z, Flockhart DA, Storniolo AM, Stearns V, Skaar TC, Hayes DF, Henry NL, and Rae JM, "CYP3A4*22 polymorphism is associated with increased exemestane concentrations in postmenopausal breast cancer patients," *Cancer Research*, vol. 76, San Antonio, TX: AACR, 2016, P5-12–05. DOI: 10.1158/1538-7445.SABCS15-P5-12-05
3. Kadakia KC, Kidwell KM, **Seewald NJ**, Synder CF, Flockhart DA, Carpenter JS, Otte JL, Hayes DF, Storniolo AM, Stearns V, and Henry NL, "Crossover from one aromatase inhibitor (AI) to another in the Exemestane and Letrozole Pharmacogenetics (ELPh) trial," *Journal of Clinical Oncology*, vol. 34, 2016, pp. 158–158. DOI: 10.1200/jco.2016.34.3_suppl.158
2. Kadakia KC, Snyder CF, Kidwell KM, **Seewald NJ**, Storniolo AM, Flockhart DA, Carpenter JS, Hayes DF, Stearns V, and Henry NL, "Associations between treatment-emergent symptoms and early discontinuation of aromatase inhibitor (AI) therapy.," *Journal of Clinical Oncology*, vol. 33, 2015, e20745–e20745. DOI: 10.1200/jco.2015.33.15_suppl.e20745
1. Mammoser AG, Weathers SPS, **Seewald NJ**, Taylor JM, and Junck L, "Primary CNS lymphoma; a review of the University of Michigan experience 2004-2013.," *Journal of Clinical Oncology*, vol. 33, 2015, e13012–e13012. DOI: 10.1200/jco.2015.33.15_suppl.e13012

Under Review

1. Gutkind S, Tormohlen KN, **Seewald NJ**, Stuart EA, Wagle P, Luo CL, and McGinty EE, "Impact of medical cannabis law on addiction treatment among adults with chronic noncancer pain and opioid use disorder," 2026, submitted.

In Preparation

2. **Seewald NJ**, *Cohen M, McGinty EE, and Stuart EA. "Ready to Roll? Whether and when to aggregate multilevel data in difference-in-differences."

1. **Seewald NJ**, *Jasper R, and Almirall D. "Sample size and timepoint tradeoffs for comparing dynamic treatment regimens in a longitudinal SMART."

Research Publications (not peer reviewed)

4. **Seewald NJ**, McGinty EE, Tormohlen K, Schmid I, and Stuart EA. "Shared control individuals in health policy evaluations with application to medical cannabis laws." arXiv: `2311.18093 [stat]`, pre-published.
3. **Seewald NJ**, Sun J, and Liao P. "MRT-SS calculator: An R shiny application for sample size calculation in micro-randomized trials." arXiv: `1609.00695`, pre-published.
2. *Moin EE, **Seewald NJ**, and Halpern SD. "Development and validation of a simple model to predict patient height," pre-published.
1. **Seewald NJ**, "Adaptive Interventions for a Dynamic and Responsive Public Health Approach," *American Journal of Public Health*, vol. 113, no. 1, pp. 37–39, 2023. DOI: `10.2105/AJPH.2022.307157`

Alternative Media

1. Morabia A (host), Chakraborty B, **Seewald NJ** (2023) "SMART and MRT - Emerging methods in public health," *AJPH Podcast*, American Public Health Association. September 13. Available at `https://youtu.be/TER4kRJpCh0`.

Teaching Experience

University of Pennsylvania Perelman School of Medicine

Course Director

2. BSTA 7000: Seminal Ideas and Controversies in Statistics — Fall 2026
   Joint with Nandita Mitra, Ph.D.

1. BSTA 6100: Biostatistical Methods for Epidemiology — Fall 2024, 2025

Guest Lecturer

2. Penn Causal Summer Institute — 2024 - Present
   "Quasi-Experimental Designs."
   Co-Directors: Nandita Mitra, Ph.D.; Eric Tchetgen-Tchetgen, Ph.D.

1. EPID 7020: Advanced Topics in Epidemiologic Research — Spring 2025, 2026
   "Quasi-Experimental Designs."
   Course Director: Ellen C. Caniglia, Ph.D.

Johns Hopkins Bloomberg School of Public Health

Co-Instructor

1. Seminar on Statistical Methods for Mental Health — Term 1, AY 2022-3
   "Promises and Pitfalls of Prediction Models in Mental Health Research"
   Co-taught with Trang Q. Nguyen, Ph.D.

Guest Lecturer

1. Seminar on Statistical Methods for Mental Health — Term 3, AY 2021-2
   "Difference in Differences with Variable Treatment Timing."
   Instructor: Elizabeth A. Stuart, Ph.D.

University of Michigan

Graduate Student Mentor

1. *STATS 250: Introduction to Statistics and Data Analysis* — AY 2019-20, AY 2020-1
   Instructor: Brenda Gunderson, Ph.D.; Jack Miller, Ph.D.

Graduate Student Instructor

3. *STATS 250: Introduction to Statistics and Data Analysis* — Spring 2018, AY 2019-20, 20-1
   Instructor: Brenda Gunderson, Ph.D.; Jack Miller, Ph.D.
2. *STATS 415: Data Mining* — Winter 2018
   Instructor: Liza Levina, Ph.D.
1. *STATS 500: Statistical Learning I: Regression* — Fall 2017
   Instructor: Brian Thelan, Ph.D.

Instructional Assistant

1. *STATS 250: Introduction to Statistics and Data Analysis* — Summer 2020
   Supervisor: Jack Miller, Ph.D.

Inter-university Consortium for Political and Social Research (ICPSR)

Teaching Assistant, Summer Program in Quantitative Methods of Social Research

2. *Multilevel Models I: Introduction and Application* — Summer 2018, Summer 2019
   Instructor: Mark Manning, Ph.D.
1. *Introduction to the R Statistical Computing Environment* — Summer 2018, Summer 2019
   Instructor: John Fox, Ph.D.

University of Notre Dame

Undergraduate Teaching Assistant

1. *BIOS 40411: Biostatistics* Spring 2012
   Instructor: Gary Lamberti, Ph.D.

Mentorship Experience

Academic Advisees

Ph.D. Students

3.	Ryan Jasper	Ph.D. Biostatistics	2025 - Present
2.	Meghan Gerety	Ph.D. Biostatistics	2025 - Present
1.	Melissa Martin Joint with Ian Barnett, Ph.D.	Ph.D. Biostatistics Senior Scientist (Biostatistics), Merck	2024 - 2026

Masters Students

7.	Daniel Goldblatt Joint with Jarcy Zee, Ph.D.	M.S. Biostatistics	2026 - Present
6.	Derek Lamb Joint with Nandita Mitra, Ph.D.	M.S. Biostatistics	2026 - Present
5.	Hao Tong, M.D.	M.S. Clinical Epidemiology	2025 - Present
4.	Ryan Jasper Joint with Jing Huang, Ph.D.	M.S. Biostatistics	2025 - 2026
3.	Luke Stewart Joint with Jinbo Chen, Ph.D.	M.S. Biostatistics	2025 - 2026
2.	Emily Moin, M.D., M.B.E.	M.S. Clinical Epidemiology	2023 - 2025
1.	Eleanor Taranto, M.D.	M.S. Clinical Epidemiology	2023 - 2025

Ph.D. Committee Membership

1.	Jimmy Kelliher Committee Chair	Ph.D. Biostatistics	2024 - Present

Biostatistics Ph.D. Lab Rotation Advisees

1. Ryan Jasper Fall 2024, Summer 2025
2. Meghan Gerety Spring 2024 - Spring 2025
   Co-advised with Rebecca Hubbard, Ph.D., in Spring 2024

Undergraduate Students

1. Mason Cohen Summer - Fall 2024

Consulting Experience

Statistical Consultant

1. *Committee for Children*, Seattle, WA — May - December 2021

Graduate Student Consultant

1. *Consulting for Statistics, Computing, and Analytics Research (CSCAR)* — Fall 2018
   University of Michigan

Presentations and Posters

Presentations at Scientific Meetings

16. Statistical considerations for interventional oncology-specific trial designs
    - Society of Interventional Oncology Annual Scientific Meeting. Savannah, GA. February 2026. *Invited*.
15. Ecological regression in health policy evaluation: A guilt-free dessert
    - Society for Epidemiologic Research Annual Meeting. Boston, MA. June 2025. *Invited.*
14. How to ask the right question: Choosing estimands for health policy research
    - International Conference on Computational and Methodological Statistics. London, United Kingdom. December 2025. *Invited.*
    - ENAR Spring Meeting. New Orleans, LA. March 2025. *Invited.*
13. Crafting relevant research: Informative and translational statistics for public policy
    - Joint Statistical Meetings. Portland, OR. August 2024. *Invited Panelist*.
12. Target trial emulation for evaluating mental health policy
    - Thomas R. Ten Have Symposium on Statistics in Mental Health. New York, NY. June 2024. *Invited.*
11. Practical guidance on whether and when to aggregate individual-level data for causal health policy evaluation
    - American Causal Inference Conference. Seattle, WA. May 2024.
10. Target trials in policy evaluation: A case study in medical cannabis laws
    - Society for Epidemiologic Research Annual Meeting. Portland, OR. June 2023. *Invited.*
9. Handling correlation in stacked difference-in-differences estimates with application to medical cannabis policy
    - International Conference on Computational and Methodological Statistics. Berlin, Germany. December 2023. *Invited.*
    - American Causal Inference Conference. Austin, TX. May 2023.
    - Joint Statistical Meetings. Washington, DC. August 2022.

8. Ready to Roll? Practical guidance on whether and when to aggregate data in health policy evaluation
   - International Conference on Health Policy Statistics. Scottsdale, AZ. January 2023.
7. Policy evaluation in the real world: Experiences with translating cutting-edge methods for an applied audience
   - American Causal Inference Conference. Berkeley, CA. May 2022.
6. Budgeting SMART: Sample size and repeated measures with a cost constraint in a longitudinal sequential, multiple-assignment randomized trial.
   - ENAR Spring Meeting. Virtual. March 2021.
5. Sample size and timepoint tradeoffs for comparing dynamic treatment regimens in a longitudinal SMART.
   - Joint Statistical Meetings. Virtual. August 2020.
4. Design considerations for comparing dynamic treatment regimens in a longitudinal SMART.
   - ENAR Spring Meeting. Virtual. March 2020.
3. Sample size considerations for comparing dynamic treatment regimes in a SMART with a longitudinal outcome.
   - International Conference on Computational and Methodological Statistics. London, United Kingdom. December 2019.
   - Joint Statistical Meetings. Denver, CO. July 2019.
2. Sample size considerations for comparing dynamic treatment regimens in a sequentially-randomized trial with a continuous longitudinal outcome.
   - Society for Clinical Trials Annual Meeting. New Orleans, LA. May 2019.
1. Sample size considerations for comparing dynamic treatment regimens in a sequential multiple-assignment randomized trial with a continuous longitudinal outcome.
   - Joint Statistical Meetings. Vancouver, BC, Canada. July 2018.
   - ENAR Spring Meeting. Atlanta, GA. March 2018.

Invited Colloquia

1. Design thinking for policy evaluation: The policy trial emulation framework
   - Department of Statistics, North Carolina State University. Raleigh, NC. April 2025.
2. Target trial emulation for evaluating health policy
   - Philadelphia Chapter, American Statistical Association. Virtual. March 2024.
3. Using individual-level data in difference-in-differences for health policy evaluation.
   - Department of Biostatistics and Data Science, Wake Forest University School of Medicine. Virtual. March 2023.
   - Department of Biostatistics, Epidemiology and Informatics, University of Pennsylvania Perelman Medical School. Virtual. March 2023.
   - Department of Public Health Sciences, Henry Ford Health. Virtual. February 2023.

- Division of Biostatistics, Northwestern University Feinberg School of Medicine. Chicago, IL. January 2023.
- Departments of Population Health Sciences and Biostatistics and Medical Informatics, University of Wisconsin-Madison. Madison, WI. January 2023.
- Department of Biostatistics, University of Colorado Anschutz Medical Campus. Virtual. January 2023.
- Johns Hopkins ALACRITY Center Research in Progress Meeting. Virtual. November 2022.

4. A brief introduction to multi-stage trials for developing dynamic treatment regimens.
   - MNeuroNetwork Lab Seminar, University of Michigan. Virtual. March 2022.
   - Working Group on Clinical Research, University of Rochester Medical Center. Virtual. October 2021.
5. Design, analysis, and sizing of sequential multiple assignment randomized trials with binary outcomes.
   - Graduate Student Statistical Topics Seminar Series, Department of Statistics, University of Michigan. September 2015.

Posters

1. "Ready to roll? Practical guidance on whether and when to aggregate data in health policy evaluation"
   - AcademyHealth Annual Research Meeting. Washington, DC. June 2022.
2. Sample size considerations for comparing dynamic treatment regimens in a sequential multiple-assignment randomized trial with a continuous longitudinal outcome.
   - Michigan Student Symposium for Interdisciplinary Statistical Sciences. Ann Arbor, MI. March 2019.
   - Thomas R. Ten Have Symposium on Statistics in Mental Health. Chicago, IL. June 2018.
3. Sample size considerations for the analysis of continuous repeated-measures outcomes in sequential multiple-assignment randomized trials.
   - Michigan Student Symposium for Interdisciplinary Statistical Sciences. Ann Arbor, MI. March 2017.
4. A SMART web-based sample size calculator.
   - Michigan Student Symposium for Interdisciplinary Statistical Sciences. Ann Arbor, MI. March 2015.
   - IMPACT Symposium III: Advances in Clinical Trial Statistics: Multiplicity Adjustment and SMARTs. Cary, NC. November 2014.

Software

1. SMARTsize: A sample size calculator for sequential, multiple-assignment, randomized trials with binary or continuous outcomes in which the primary aim is to compare two embedded dynamic treatment regimes.
   https://nseewald1.shinyapps.io/SMARTsize/

2. MRT-SS Calculator: A sample size calculator for micro-randomized trials in which the primary aim is to detect the proximal effect of providing an intervention.
   `https://pengliao.shinyapps.io/mrt-calculator/`

Professional Service

Public Training Materials

1. "Target Trial Emulation for Evaluating Health Policy."
   Johns Hopkins ALACRITY Center for Health and Longetivity in Mental Illness
   `https://www.youtube.com/watch?v=DAXfp8X9ba8`
2. "Building Effective Adaptive Interventions in Mental Health Services Research."
   Johns Hopkins ALACRITY Center for Health and Longevity in Mental Illness
   `https://youtube.com/playlist?list=PLj4-nn-p-zv1o-cR9jiITTRfgkwmyqjoD`

Workshop Design and Facilitation

1. Society of Interventional Oncology Clinical Trials Methodology Workshop
   - Steering Committee member and biostatistics faculty. Philadelphia, PA. May 19-23, 2026.
2. "SMART Designs for Developing Adaptive Interventions"
   - Statistical Horizons, Virtual. July 8-11, 2025.
   - Statistical Horizons, Virtual. August 19-23, 2024.
3. "Embedding Qualitative and Quantitative Methods for Policy Evaluation"
   - Invited pre-conference workshop: 15th International Conference on Health Policy Statistics, San Diego, CA. January 6, 2024. *Co-designer and facilitator with Beth McGinty, Ph.D. and Eli Ben-Michael, Ph.D.*
4. "Introduction to Adaptive Interventions and Sequential Multiple Assignment Randomized Trials"
   - Invited pre-conference workshop: Society of Behavioral Medicine Annual Meeting, Baltimore, MD. April 6, 2022. *Co-designer and facilitator with Ahnalee Brincks, Ph.D., and Shawna N. Smith, Ph.D.*
5. "Building Just-in-Time Adaptive Interventions."
   - Center for Dissemination and Implementation Science Summer Institute, University of Illinois Chicago. Virtual. October 25-26, 2021. *Small group expert facilitator*.
6. "Using SMART Design in Responsive Survey Design."
   - Responsive Survey Design: A Research Education Program. Institute for Social Research, University of Michigan. June 19, 2019. *Co-designer and facilitator with Ahnalee Brincks, Ph.D.*
7. "Getting SMART about Adaptive Interventions in Education."
   - Institute of Education Sciences-funded training institute at the University of Michigan. March 11-14, 2019. *Co-designer and facilitator with d^3lab* `d3lab.isr.umich.edu`.
8. "Getting SMART: Experimental Design and Analysis Methods for Developing Adaptive Interventions"

- University of California, San Francisco. May 19, 2015. *Co-facilitator with Inbal Nahum-Shani, Ph.D.*

9. “Experimental Design and Analysis Methods for Developing Adaptive Interventions: Getting SMART”
   - Methodology Center Summer Institute. Ann Arbor, MI. June 19-20, 2014. *Small group facilitator*.

Peer Review and Editorial Activities

Editorial

2. Associate Editor, American Journal of Epidemiology. 2026 - Present.
1. Guest Academic Editor, PLOS Medicine. 2025.

Ad hoc Reviewer, Academic Journals

American Journal of Epidemiology · American Journal of Public Health · Annals of Applied Statistics · Behavior Research Methods · Biometrics · BMJ · Clinical Trials · Contemporary Clinical Trials · Data Science in Science · Frontiers in Digital Health · Health Services Research · JAMA Health Forum · JAMA Network Open · JAMA Pediatrics · Journal of the American Statistical Association (Theory and Methods) · Journal of Causal Inference · Pharmacoepidemiology and Drug Safety · PLOS Medicine · Psychological Methods · R Journal · Statistics and Public Policy · Statistics in Medicine · Value in Health

Grant Review

3. Scientist Reviewer (methodologist), PCORI Merit Review Panel. Patient-Centered Outcomes Research Institute (PCORI). 2026.
2. Panel Member, Pharmacotherapy of Substance Use Disorder (PSUD). National Institutes of Health (NIH). 2026.
1. Subject Matter Expert reviewer of Draft Final Report. Patient-Centered Outcomes Research Institute (PCORI). 2025.

Departmental Service

8. Member, Biostatistics Faculty Recruitment Committee. Department of Biostatistics, Epidemiology, and Informatics, University of Pennsylvania. 2025 - Present.
7. Member, Biostatistics Seminar Committee. Department of Biostatistics, Epidemiology, and Informatics, University of Pennsylvania. 2025 - Present.
6. Member, Epidemiology Qualifying Exam Committee. Graduate Group in Epidemiology and Biostatistics, University of Pennsylvania. 2025 - Present.
5. Co-Chair, Junior Faculty Committee. Department of Biostatistics, Epidemiology, and Informatics, University of Pennsylvania. 2023 - Present.
4. Member, Biostatistics Student Recruitment Committee. Graduate Group in Epidemiology and Biostatistics, University of Pennsylvania. 2023 - Present.

3. Student Representative, Faculty Diversity, Equity, and Inclusion Committee. Department of Statistics, University of Michigan. AY 2020-2021.

2. Co-Lead, Graduate Student Instructor Foundations Colloquium. Department of Statistics, University of Michigan. AY 2020-2021.

1. Steering Committee Member, Graduate Student Diversity, Equity, and Inclusion Working Group. Department of Statistics, University of Michigan. 2020-2021.

Professional Memberships

American Statistical Association (2015 - present)
International Biometrics Society, Eastern North American Region (ENAR) (2015 - present)
Society for Causal Inference (2023 - present)
Society for Clinical Trials (2015 - 2021)
Society of Industrial and Applied Mathematics (SIAM) (2020-2021)
Society of Behavioral Medicine (2022 - 2023)

Organizational Activities

11. Member, Program Committee, 2027 American Causal Inference Conference. Society for Causal Inference. 2026-2027.

10. Member, Outreach Committee, 16th International Conference on Health Policy Statistics. 2026.

9. Abstract Concurrent Session Organizer, 2026 American Causal Inference Conference. Society for Causal Inference. 2025-2026.

8. Member, Program Committee, 18th Annual Conference on Statistical Issues in Clinical Trials. University of Pennsylvania. 2025-2026.

7. Member, Program Committee, 2026 American Causal Inference Conference. Society for Causal Inference. 2025-2026.

6. Session Chair, ENAR Spring Meeting. March 2025.

5. Chair, Diversity Equity and Inclusion Committee. Society for Causal Inference. 2025.

4. Member, Program Committee, 17th Annual Conference on Statistical Issues in Clinical Trials. University of Pennsylvania. 2024-2025.

3. Member, Outreach Committee, 15th International Conference on Health Policy Statistics. 2024.

2. Session Chair, 14th International Conference on Health Policy Statistics. January 2023.

1. Session Chair, Joint Statistical Meetings. August 2022.

Professional Development

4. U-M Graduate Teacher Certificate. Rackham Graduate School and Center for Research on Learning and Teaching, University of Michigan. Completed April 2021. crlt.umich.edu/um.gtc

3. Inclusive STEM Teaching Project. Completed December 2020. inclusivestemteaching.org

2. Preparing to Teach: Workshop to prepare graduate students for a role as undergraduate faculty responsible for teaching statistics and data science. May 2020. preparingtoteach.org

1. Professional Development Diversity, Equity, and Inclusion Certificate. Rackham Graduate School, University of Michigan. Completed April 2020. rackham.umich.edu

Research Grants

Funded

1. Impact of U.S. State Recreational Cannabis Laws on Opioid Prescribing for Chronic Non-Cancer Pain and Statistical Methods for Informative Policy Adoption

   Role: Principal Investigator
   Funding Source: Thomas B. McCabe and Jeannette E. Laws McCabe Fund
   `https://www.med.upenn.edu/evdresearch/mccabe-fund.html`
   Funding Level: Pilot Award, 7/1/24 - 12/31/25: $36,347